

#Step 1 → Gathering Information Necessary for Linear Interpolation

- Variables Needed
 - excess_temp #Has the user input the excess temp for me to use
 - Ax, Ay #Data points at point A (x, y)
 - Bx, By #Data points at point B (x, y)
 - Cx, Cy #Data points at point C (x, y)
 - Dx, Dy #Data points at point D (x, y)
 - Ex, Ey #Data points at point E (x, y)
 - $m = \frac{\log(_y - _y)}{\log(_x - _x)}$
 - Determined by the region of the graph
 - $\text{heat_flux} = _y(\text{excess_temp}/_x)^m$
 - Determined by the region of the graph

#Step 2 → Determine the Region of the Graph The Point is Located

- I'll use the output for the excess temperature to determine where on the graph the point is
 - If $1 \leq \text{excess_temperature} < 5$
 - Free Convection section (use points A and B)
 - If $5 \leq \text{excess_temp} < 30$
 - Nucleate section (use point B and C)
 - If $30 \leq \text{excess_temp} < 120$
 - Transition section (use points C and D)
 - If $120 \leq \text{excess_temp}$
 - Film sections (use points D and E)
- Based off where on the graph I am I will use the corresponding slope and linear interpolation equation to get a y-value on the corresponding line segment
- Lastly if the excess_temperature is less than 1 or greater than 1200
 - print("Surface heat flux is not available")

#Step 3 → Calculate the Heat Flux

- I'll calculate the heat_flux using the corresponding equation and print it out
- print(f'The surface heat flux is approximately {heat_flux} W/m^2')

Test Cases

1. Typical: 3
 - a. Output: 3347 W/m²
2. Typical: 10
 - a. Output: 55828 W/m²
3. Typical: 49
 - a. Output: 352198 W/m²
4. Edge: 0
 - a. Output: Surface heat flux is not available
5. Edge: 1901
 - a. Output: Surface heat flux is not available
6. Edge: (-1)
 - a. Output: Surface heat flux is not available
7. Typical: 87
 - a. Output: 64628 W/m²
8. Typical: 1
 - a. Output: 685 W/m²
9. Typical: 1200
 - a. Output: 1500000 W/m²
10. Typical: 198
 - a. Output: 60906 W/m²